Hacking Electronic Music

Live coding in the browser with Mercury

Timo Hoogland



General info

length: +/- 3 hours (longer or shorter possible)

participants: Max. 20 people (preferably +/- 15)

ages: 16+

skills: no programming or musical experience required!

computer: Bring your own laptop and headphones

language: English/Dutch (depending on the participants)

system requirements:

- Mac 10.11+ / Windows 10 / Linux:
  - Intel Core i5 processor (or faster recommended)

  - 4 GB RAM (8 GB or more recommended)
  - Firefox or Chromium browser (Firefox, LibreWolf, Brave, Arc)

websites:

- https://timohoogland.com
- https://mercury-playground.cc

Workshop description

Join this crash-course into the world of live coding electronic music. No previous knowledge of programming is required, only an interest to express yourself musically! During this workshop you will learn to create electronic music with computercode and algorithms. For this we use the programming environment Mercury. You will program instructions, and the computer will translate those to beats, synthesizers and other crazy sounds. You can adjust the code while the music is playing, and hear the sound change. This is the beginning of live coding! At the end of the workshop you have been introduced to technologies used in electronic music production and you will have experienced what it means to use algorithms to compose music and sound. We'll also look briefly at the larger live coding scene and what other things you could explore in the future.

In-depth description of Mercury

Mercury is a human-made, beginner-friendly and human-readable language that focusses on quick expression of code for composing, performing and communicating live music. Mercury is a good environment to get introduced to the worlds of live coding, creative coding, algorithmic composition, electronic music and of course the Algorave scene! In Mercury, all elements of the language are designed around making code more accessible and less obfuscating for you (the performer), but also the audience. This motivation stretches down to the coding style itself which uses clear descriptive names for functions and a clear syntax. Furthermore the environment runs in the browser so it is available to you at all times. Mercury attempts to provide the artist-programmer with an extensive library of algorithms to generate sounds, sequences and music.

Biography

Timo is a digital artist, live coder, drummer and music technology educator from the Netherlands. He livecodes electronic music and develops algorithmic audiovisual compositions and installations. Timo graduated from the Masters of Music at the HKU University of the Arts Utrecht in 2019, where he developed the livecoding tool Mercury. He organizes meetups, workshops and algoraves as part of Netherlands Coding Live. He teaches creative coding at HKU Music Technology and Conservatory of Amsterdam.

Socials

- https://www.timohoogland.com
- https://www.instagram.com/tmhglnd
- https://social.toplap.org/@tmhglnd
- https://www.youtube.com/tmhglnd

- https://timohoogland.bandcamp.com/

Hashtags

#electronicmusic #algorithmicmusic #creativecoding #livecoding #algorithmiccomposition #sound-design #digitalart #digitalculture #diyprogramming #algorave #performanceart

Goals

Starter Goals

- Learn to play a sample
- Change the timing of the played sound
- Make lists as rhythms and use probabilities
- Generate lists with functions

Intermediate Goals

- Learn to play a synth
- Make lists as melodies and rhythms for the synth
- Generate lists of melodies and rhythms with functions

Advanced Goals

- Combine functions to generate more complex melodic and rhythmical structures
- Add effects for sound designing
- Sequence parameters of effects and other functions
- Include visuals coded with Hydra
- Control synths with external devices via OSC
- Sequence other applications via MIDI output

Schedule (by approximation)

15 minutes :

- Introduction to livecoding and showcase some works

30 minutes :

- Syntax Explanation
- Work on starter goals

30 minutes :

- Play and explore possibilities with rhythms

15 minutes :

- Break

30 minutes :

- Work on Intermediate goals

45 minutes :

- Play and explore!
- Individual help/questions
- Optional move to Advanced goals

15 minutes :

- Wrap up
- Participants show their creations

Technical Requirements

for more details see the Workshop Technical Rider

Sound:

- Speakers and Mixer (PA) with mini-jack

Visual:

- Projector or screen with HDMI connection
- Flipover board or whiteboard and black (or colored) markers

Participants:

- Tables
- Chairs
- Power sockets on table

Consumptions:

- Coffee / Tea / Water / (Soda)
- Snack or lunch (depending on length and time of workshop)

Thanks to

- Mercury has been granted funding from **Creative Industries Fund NL**
- Mercury is sponsored by **Creative Coding Utrecht**

Credits

- Photo by Indiara de Benedetto